Computer Networks (Class-Notes)

- LAN
- MAN
- WAN

Tuesday
28 July, 2026

- Internet & computer network difference.
- OSI (7-layers) → Theoretical framework

open source interconnection

- Physical layer
- Data link "
- Network "

} Hardware Layers

- Transport " → Interconnecting Layer
- Session "
- Presentation layer
- Application "

} Software Layers

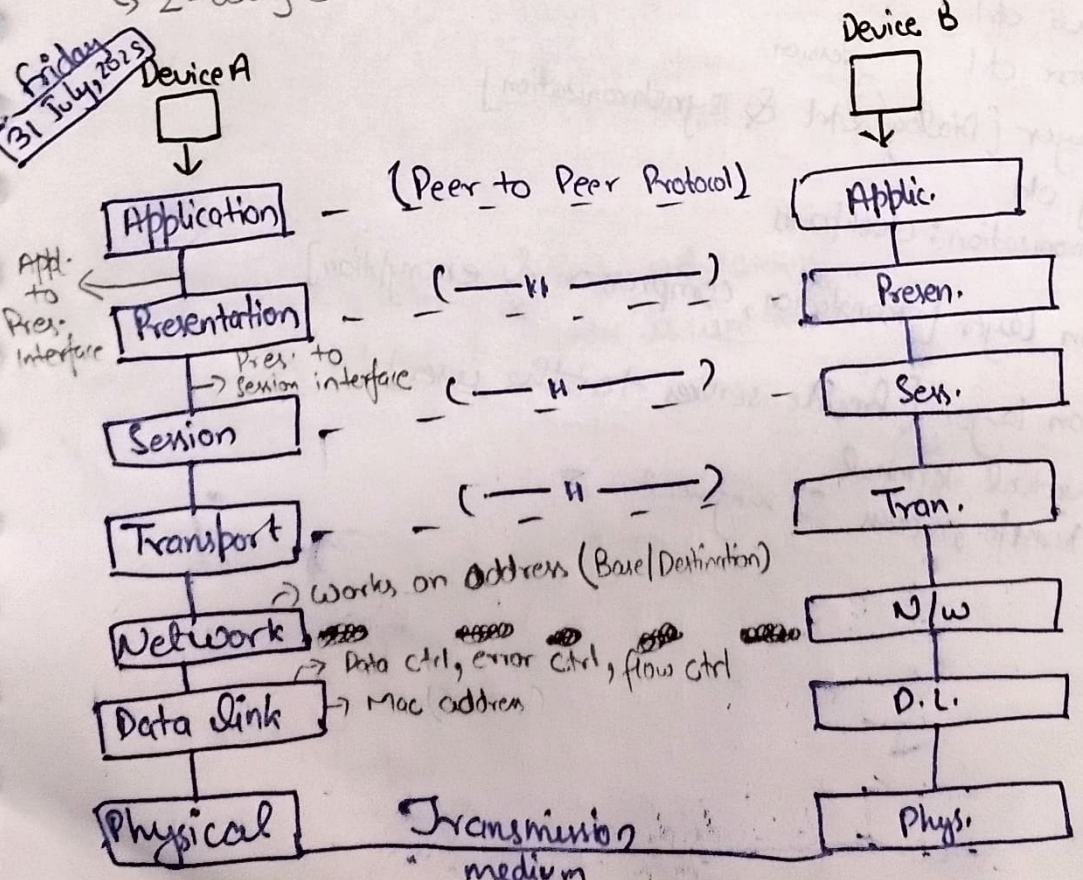
Types of mediums for data transmission??

- Guided
  - Twisted Pair
  - Coaxial
  - Fibre optics
- Unguided
  - Radiowaves
  - Microwaves
  - Infrared

Transmission Modes

- Simplex (One way) → Bus topology
- Half-duplex → Two-way (1 at a time)
- Full - "
  - 2-way Simultaneous

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31 July, 2026



Friday
7 Aug, 2020

P.P.T.

⇒ Presentation layer (transmits bits over medium)
→ ~~Cont~~ Properties

⇒ Data link layer (deliver frame from hop(node) to hop)
→ Properties
- Framing
- Physical addressing
- Flow ctrl
- Access "
- Error "
- hop to hop delivery

⇒ N/w layer (Packet delivery from S to R)
→ Properties
- Logical addressing
- Routing
- S to destination delivery

⇒ Transport layer (Delivery of msg from 1 process to another) → unique port no.
→ Properties
- Service point addressing / port addressing
- Segmentation & reassembly
- Connection ctrl
- Flow ctrl
- Error ctrl

⇒ Session layer [Dialog ctrl & synchronization] → session
- Dialog ctrl
- Synchronization : checkpoints

⇒ Presentation layer [translation, compression & encryption]

⇒ Application layer [Provide services to the user]
- N/w virtual terminal
- File transfer, access & mngment

Monday
10 Aug 2026

TCP/IP Protocol Suite

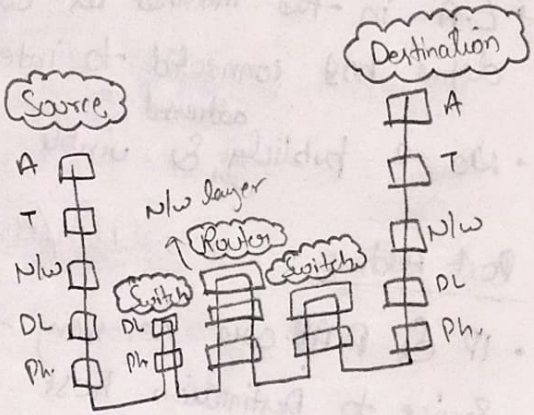
→ Used in internet today
→ Set of protocols organized in diff. layers.
→ Hierarchical prot. made up of interactive modules.
→ Each module provides specific functionality.
→ Defined as 4 s/w layers built upon H/w. → Host-to-n/w, Internet, Transport, Application → also d/s n/w interface
→ When compared to OSI model, 5 layers → Physical, Data link, N/w, Transport, Application → H/w devices, N/w Interface, Internet

Applic.
SMTP, FTP, HTTP, DNS, TELNET
Specific Addresses
→ Appl. + Presentation + Session

Transp.
SCTP, TCP, UDP
Port Addresses

Internet
ICMP, IGMP
IP
RARP, ARP
→ N/w Layer
Logical Add.

DLL, Phys. Layer
Protocols delivered by the underlying n/w (host to n/w)
Physical Add
→ Data link + Physical layer



Switch: Increase the ~~da~~ strength of damping signal

⇒ Duty of Ap., T, N/w layers is end-to-end
⇒ Domain of duty of top 3 layers is internet

...
...
...

Address → Phys.
→ Logical
→ Port
→ Specific

Physical Address

- Also known as Link add., is the add. of node as defined by its LAN/WAN
- Lowest-lvl addr.
- Have authority over n/w
- Included in the frame used by DLL
- Size & format of these vary depending on the n/w

Logical Address

- Necessary for universal comm. that are independent of underlying phy. n/w
- L.A. in the internet is currently a 32-bit add. that can uniquely define host connected to internet
- No 2 publicly addressed & visible hosts can have same IP add.

Port Address

192.168.1.1:8080

- IP & P.Add. are necessary for a quantity of data to travel from Source to Destination host.
- End objective of internet comm. is a process comm. with another process.
- Port add. in TCP/IP is 16-bits in length.

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11 Aug, 2020

N/w Layer

At this layer (internetwork layer), TCP/IP supports

IP: Unreliable & connectionless, a best-effort delivery service
→ Assumes unreliability of underlying layers & does its best to get transmission through to its destination
→ transports packets cls datagrams, transported seperately
→ Datagrams can travel along diff. routes

Address Resolution Prot. (ARP): used to associate L.A. with Ph. add.

→ Used to find P.A. when internet add. is known

Reverse ARP: Allows host to discover its internet add.

→ Used when computer connected 1st time or when diskless computer is booted

Signals

→ Single valued func. of time
→ time plays independent variable
→ Means at every instant of time, signal has unique value

Types

- Continuous, discrete time
- Analog, digital → discrete digital
- Periodic, aperiodic
- finite, infinite length
- Causal $f(t)=0 \ \forall t<0$, anticausal $f(t)\neq 0 \ \forall t\geq 0$, noncausal $\exists t_1<0 : f(t)=0$
- Even $f(t)=f(-t)$, odd $f(t)=-f(-t)$
- Deterministic & random
  - No uncertainty
  - Defined exactly by mathematical formula
  - {Non-deterministic / Probabilistic}

Electromagnetic Spectrum

- Electric waves
- Magnetic "

10^{24} [ν(Hz)] — γ-rays | X-rays | UV | Visible | IR | Microwave | FM | AM | long radio waves — 10^{0} … 10^{8}

10^{-16} [λ(m)]

Radiowaves

→ IR applications

Comm. System

Classification based on

- Direc. of comm.
  - Simplex
  - Half-duplex
  - Full- "
- Nature of info signal
  - Analog
  - Digital
- Techniques of transmission
  - Baseband transmission
  - Comm. using modulation

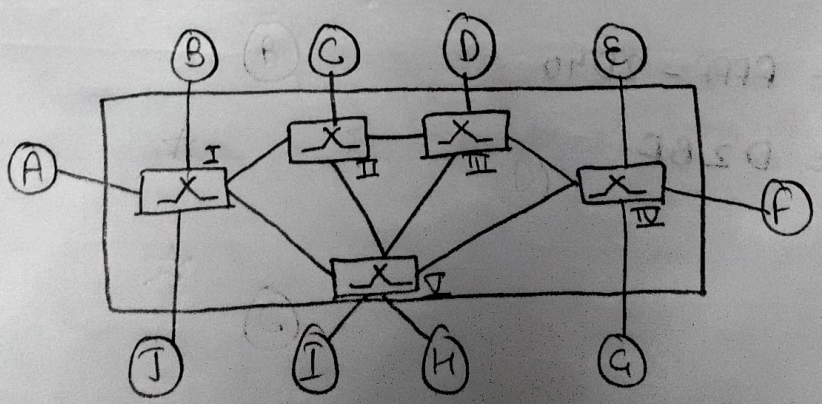
Tuesday
1 Sept, 2015

TUTORIAL - 1

1.) Describe the need for switching & define a switch in C.N.
2.) Compare circuit switch network with packet switching n/w.
3.) Discuss the 2 approaches to packet switching.
4.) " diff. kinds of networking devices.

1.) When we deal with very large networks, most of the topologies become impractical. For example, the mesh or star topologies require too much infrastructure which are cost-efficient. Some other topologies, such as bus, are ruled out because of very large distances between the devices (large network case). Here arises the need for switching.

Switches are the interlinked devices (or nodes) in a network that create a temporary connection between 2 or more devices linked with them. Some switches are connected to the end systems while others are used only for routing.



Here, A, B, C, D.... are end systems & I, II, III, IV & V are switches

3.) The two approaches to packet-switching are discussed below:

(i) Datagram approach

In this approach, each packet (or datagram) is treated independently of others. Each switch has a routing table in which the destination addresses & the corresponding forwarding output ports are recorded. When the switch receives the packet, it examines the destination address & consults the routing table to find corresponding port to forward the packet.

Destination Address	o/p port
1232	1
4150	2
⋮	⋮
9130	3

→ Routing table

(ii) Virtual-Circuit approach

In this approach, the source & destination need to go through 3 phases: setup, data transfer, & teardown.

- In the setup phase, the source & destination use their global addresses to help switches make table entries for the connection.
- In the data-transfer phase, the frames are transferred through switches. The process creates a virtual-circuit.
- In the teardown phase, the source & destination inform the switches to delete the corresponding entry.